Blockchain in Finance

Adoption and application

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Disclaimer

I am no finance expert

- Many largely announced case studies
- Few academic sources
- These slides are based on thorough but not exhaustive reading.

Blockchain and finance

- High expectations in 2015, 2016.

> “Bitcoin is not just a minor incremental change, it’s not a payments network. Bitcoin is one of the most fundamental transformations on the basis of money,”
>
> Andreas Antonopoulos during a WIRED Money 2015 summit.
> https://www.coinspeaker.com/bitcoin-is-punk-rock-you-cant-control-it-says-andreas-m-antonopoulos-10612/

> “Rather than stay at the margins of the finance industry, blockchain will become the beating heart of it,”
>
> Giancarlo Bruno, Head of Financial Services Industries, World Economic Forum. 2016
> https://www.weforum.org/media/blockchain-will-become-beating-heart-of-the-global-financial-system/

- Where, how and why is blockchain or similar technology used today?

Blockchain and finance

- Potential of tokenisation:
  - Atomic settlement
  - 24/7
  - Transparency
  - Avoid manual handling

“Tokenisation - the digital representation of claims on programmable ledgers”

DLT - distributed ledger technology

“*Programmability does not strictly require DLT –*
many functions can be implemented on traditional infrastructures –
but combining DLT with tokenisation can deliver notable benefits. “

Source: BIS (An bank connecting national central banks)
Report 2025: https://www.bis.org/publications/next-generation-monetary-financial-system.pdf,
Report 2026: https://www.bis.org/publications/anchoring-trust-money.pdf

Blockchain and finance

Examples

- Bakong Payment system - Cambodia
- Pontes - EU
- mBridge - BIS
- Commercial platforms
- Norway?

Some Concepts

Concepts

- Central bank money vs commercial money
  cash vs bank account balance

- CBDC
  digital central bank money

- Wholesale / Retail
  used by banks and financial institutions (wholesale) or households and businesses (retail)

Bakong

Bakong

What

- Payment and settlement system run by Cambodia’s central bank
- Banks and payment providers can join
- Users can send money using the Bakong app
- Can pay online, to a friend or in a store
- Unbanked users can send limited funds
- Banks perform KYC (know your customer)

Bakong

What

- Payment and settlement system run by Cambodia’s central bank
- Banks and payment providers can join
- Users can send money using the Bakong app
- Can pay online, to a friend or in a store
- Unbanked users can send limited funds
- Can send dollar and Khmer riel
- Banks perform KYC (know your customer)

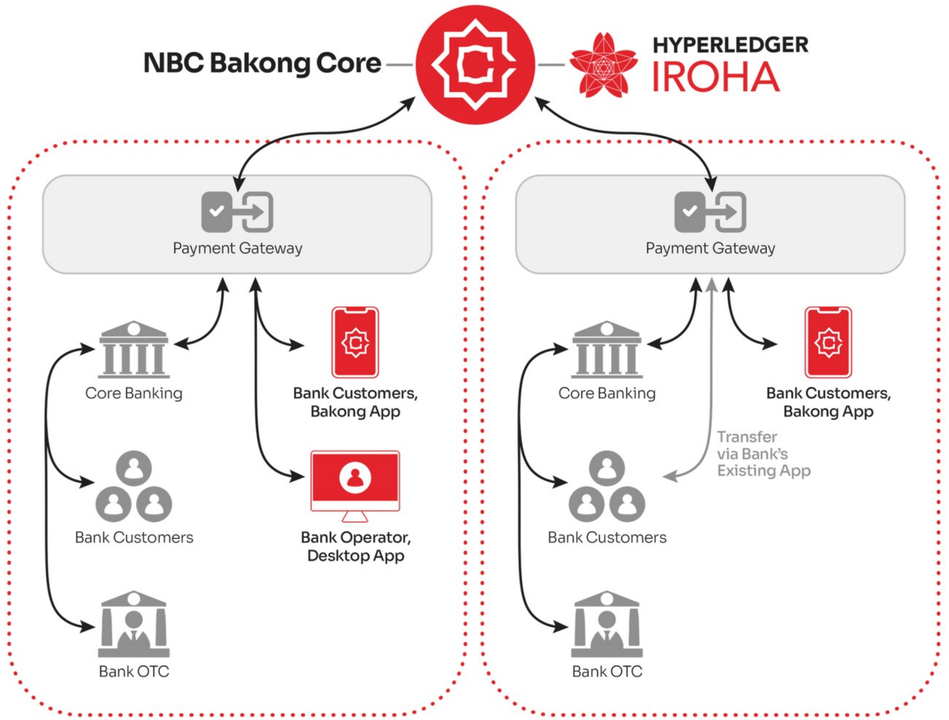


Image: Sorimatsu

Bakong

How

- A blockchain (Hyperledger Iroha) records transactions
- Multiple blockchain nodes run at the national bank
- All transactions are recorded on the blockchain
- Transactions carry no personal information
- 2.000tps, 5sec latency

Source https://bakong.nbc.gov.kh/download/NBC_BAKONG_White_Paper.pdf

Bakong

Questions

- Is is decentralised?
- Censorship resistance?
- Payment only, no smart contracts.

Source https://bakong.nbc.gov.kh/download/NBC_BAKONG_White_Paper.pdf

Pontes

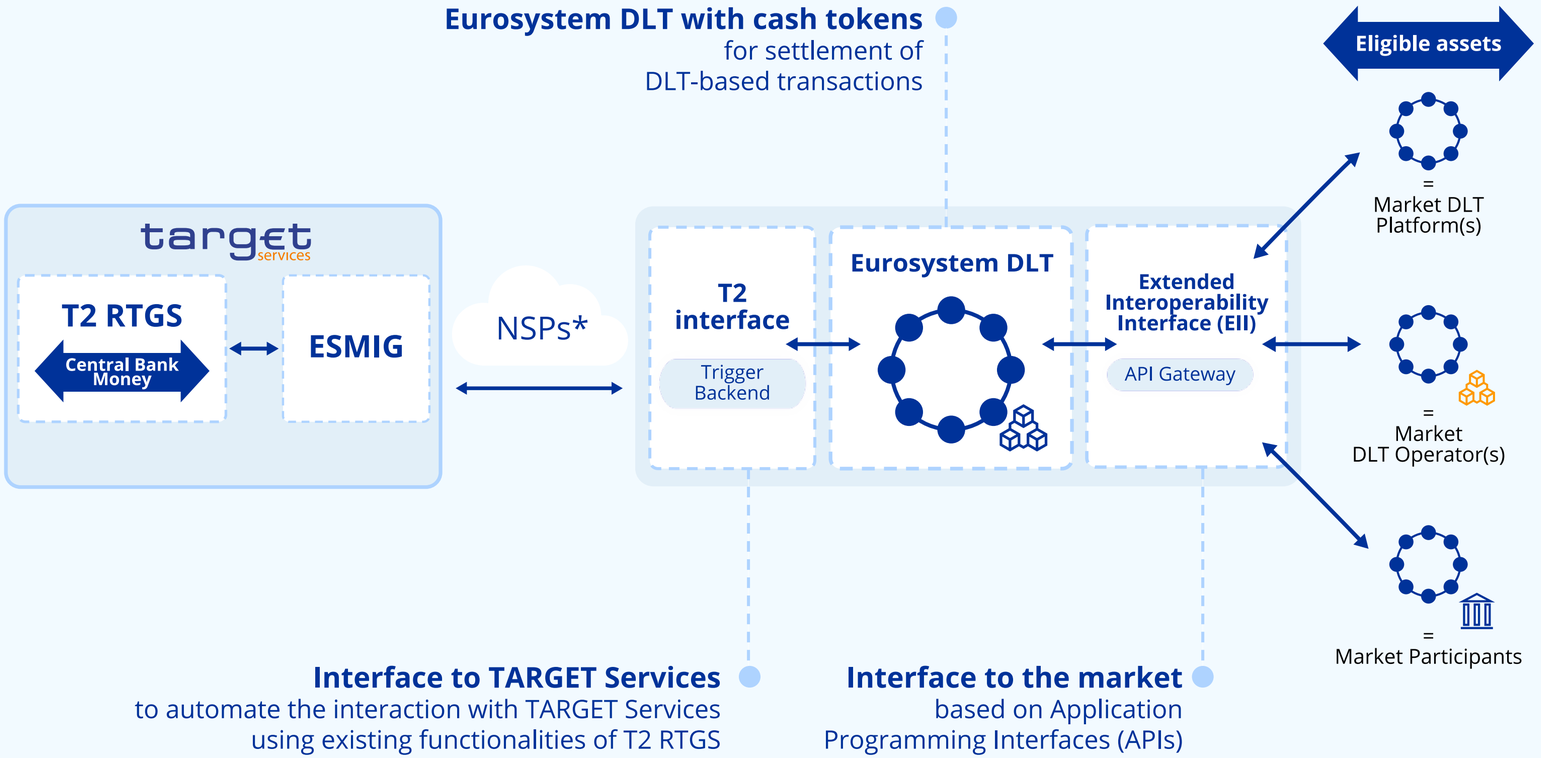
Pontes

What

- Pontes is a Eurosystem service that lets DLT trading platforms settle in central bank money
- A bridge between DLTs and Central bank money
- Pay for tokenised bonds and shared with safe cash
- Only for banks and financial institutions
- Launched 21. September 2026
- Open Monday to Friday 9 to 16.

Pontes

What



*Network Service Providers

The infographic refers to the initial launch of Pontes.

Source: Eurosystem

Pontes

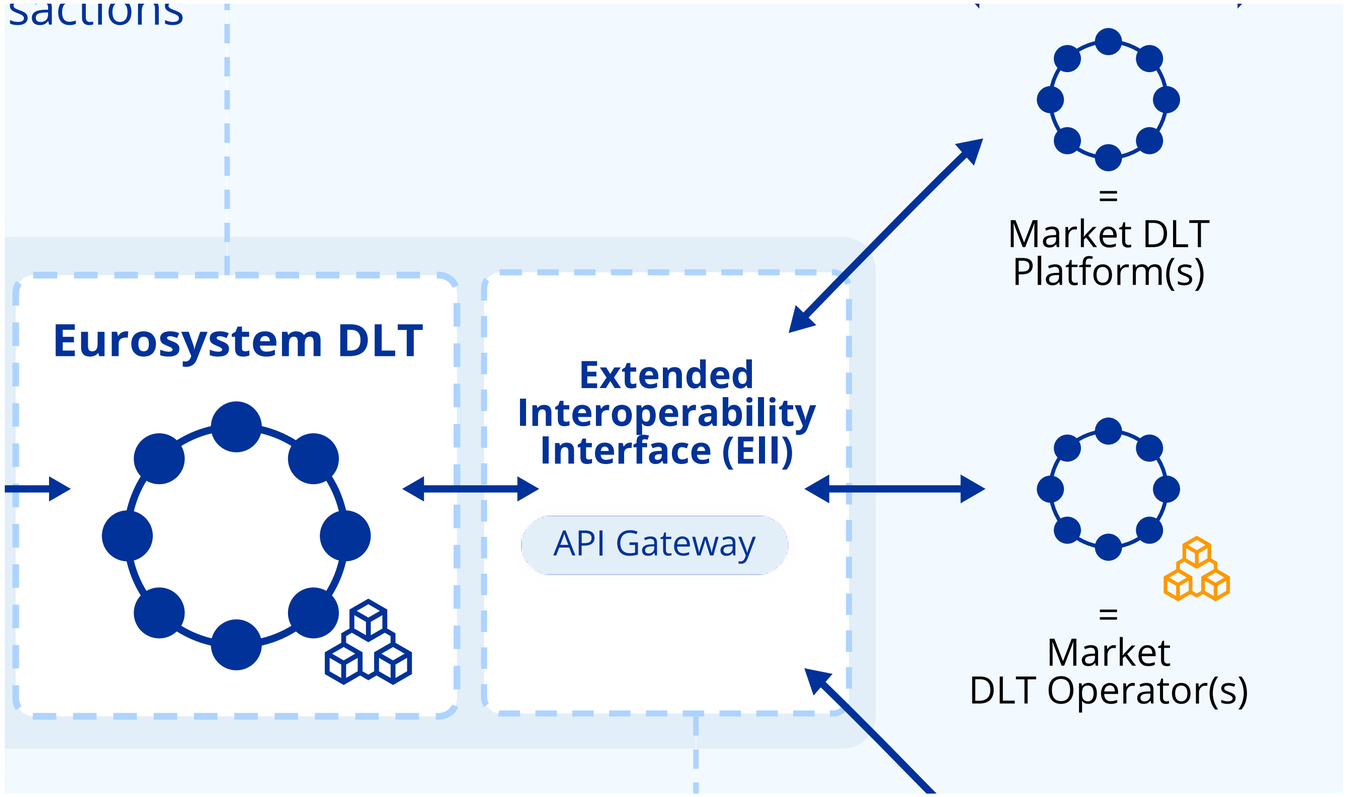
What

- Pontes blockchain:
- Hyperledger Fabric
  - Central Bank Nodes (one per NCB) – validate, sign, and store transactions and manage participant data.
  - ECB Issuer Node – issues and redeems cash tokens

Pontes

Hash-lock protocol

- Exchange asset on Market DLT for payment on Eurosystem DLT
- Register trade with EII
  returns
  cancellation hash
  execution hash



- Payment made on Eurosys DLT -> EII releases execution preimage
- Timeout without payment -> EII releases cancellation presage
- EII is trusted Oracle

Pontes

Questions

- How decentralised is the Eurosystem DLT in pontes?
- Does it give censorship resistance?
- How does the Hash-lock compare to a bridge?

mBridge

mBridge

What

> “Project mBridge experiments with cross-border payments using a common platform based on distributed ledger technology (DLT) upon which multiple central banks can issue and exchange their respective central bank digital currencies (multi-CBDCs).”

https://www.bis.org/publications/project-mbridge-connecting-economies-through-cbdc.pdf

mBridge

What

“Cross-border payments are typically made through a global network of correspondent banks involving multiple intermediaries that are fragmented across different time zones and operating hours.
… due to duplicated processes and steps in the correspondent banking chain, cross-border payments exhibit high costs, low speed, operational complexities, limited access and low transparency.”

“Furthermore, cross-border transactions are often settled in a handful of dominant currencies and FX trading involving non-dominant currency pairs remains limited.

https://www.bis.org/publications/project-mbridge-connecting-economies-through-cbdc.pdf

mBridge

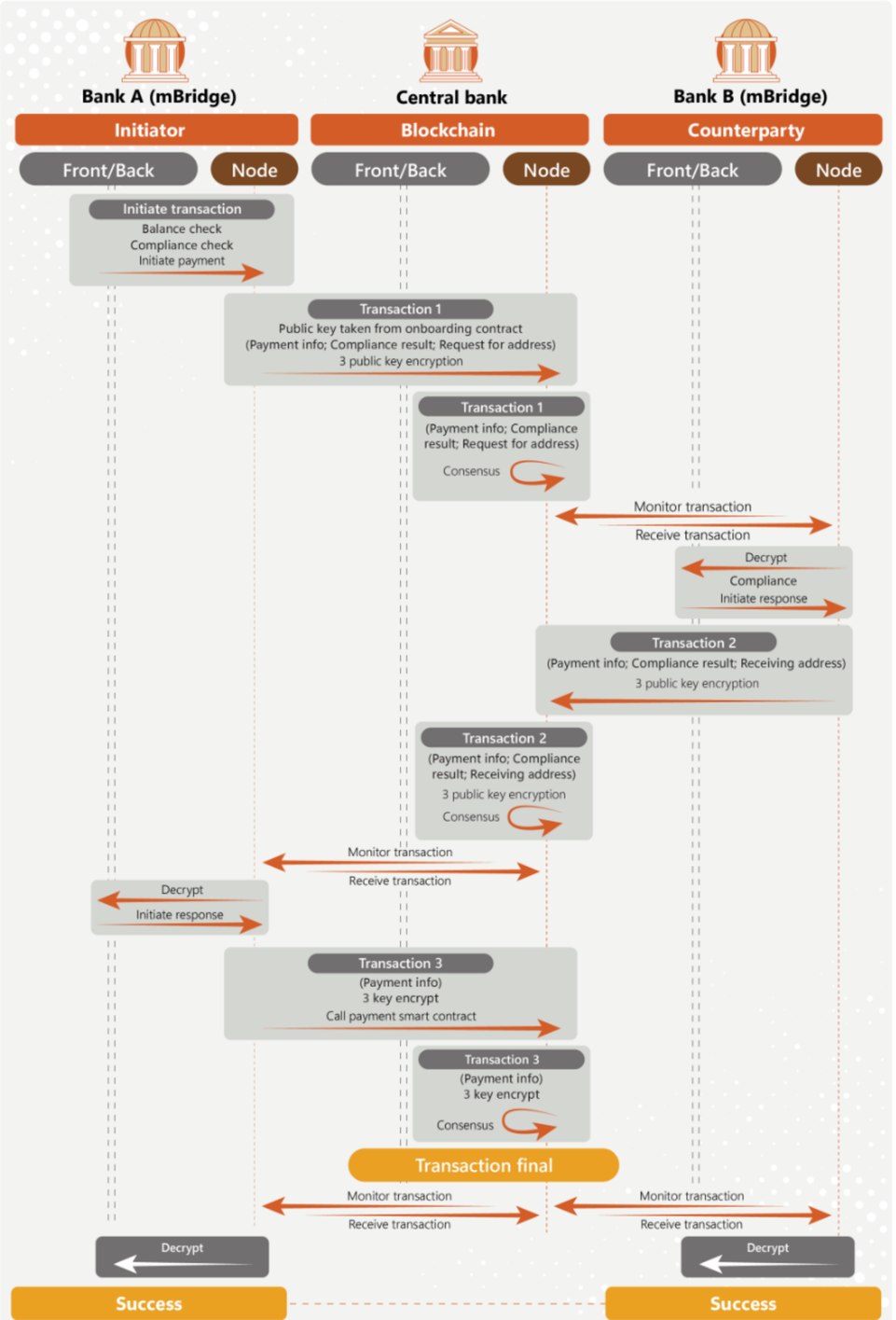
How

- mBL mBridge ledger
  - Private permissioned blockchain
  - Central banks each run a validating node
  - Uses HotStuff+, or Dashing consensus protocol
  - Privacy:
    Payer and payee identities, amount, … are only visible to
    Payer, payee, payer and payees central banks and currency issuer.

mBridge

Example

- mBL mBridge ledger
  - Encryption
    (for whom?)
  - Communication on chain



mBridge

Status

- 2021 - 2024 pilot including BIS and National bank of Thailand, Hong Kong, China
- 2024 BIS stops participation
- Still used?

mBridge

Questions

- How decentralised is it?
- Does it give censorship resistance?

Commercial platforms

Commercial platforms

What

- Payment vs payment
- “Currency exchange”
- But also DvP: Delivery vs. Payment

Commercial platforms

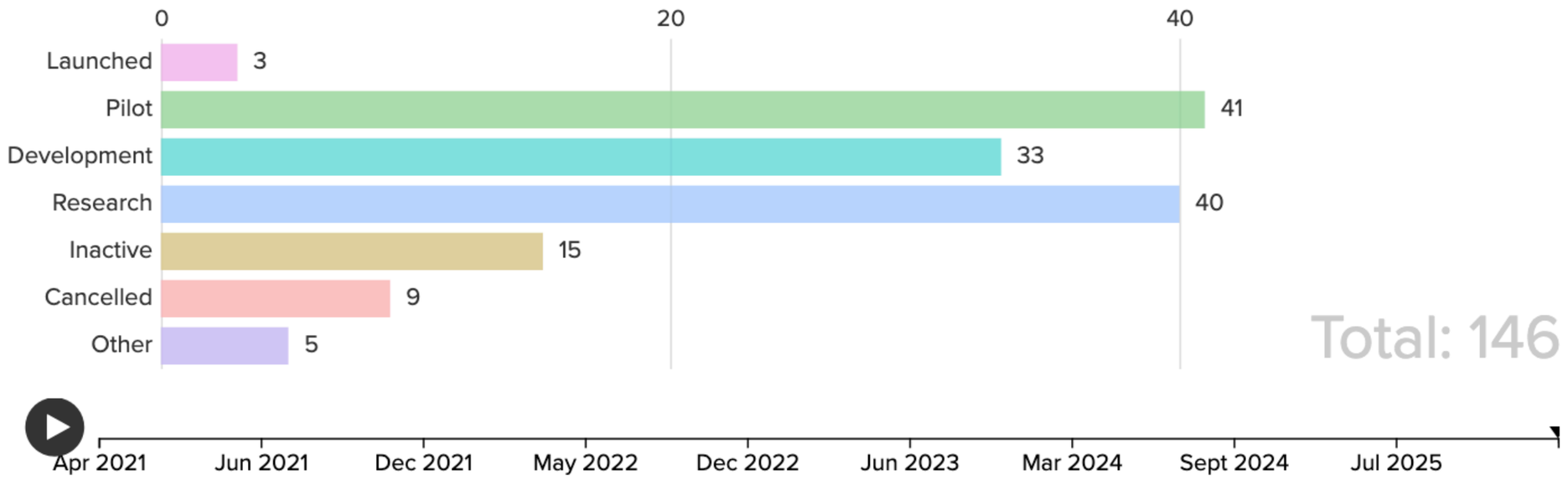
Examples

- JP Morgan Kinexys: https://www.jpmorgan.com/kinexys/index
- Partior: https://partior.com/

- Single or Multi bank organisation
- Who is running the ledger?

Other projects

- Check https://www.atlanticcouncil.org/cbdctracker/



Norway

> “CBDC is probably not the most suitable measure for promoting payment system innovation, strengthening preparedness or in another way contribute to a more efficient and secure payment system at this stage.”

- Source: Norges bank CBDC final report 2026

Takeaway 1

- Adoption exists but is limited
- Blockchain used more for added security than for decentralisation
- Atomic swap and cross pratform connections are similar to bridges and atomic swap smart contracts, but may rely on trusted third parties.

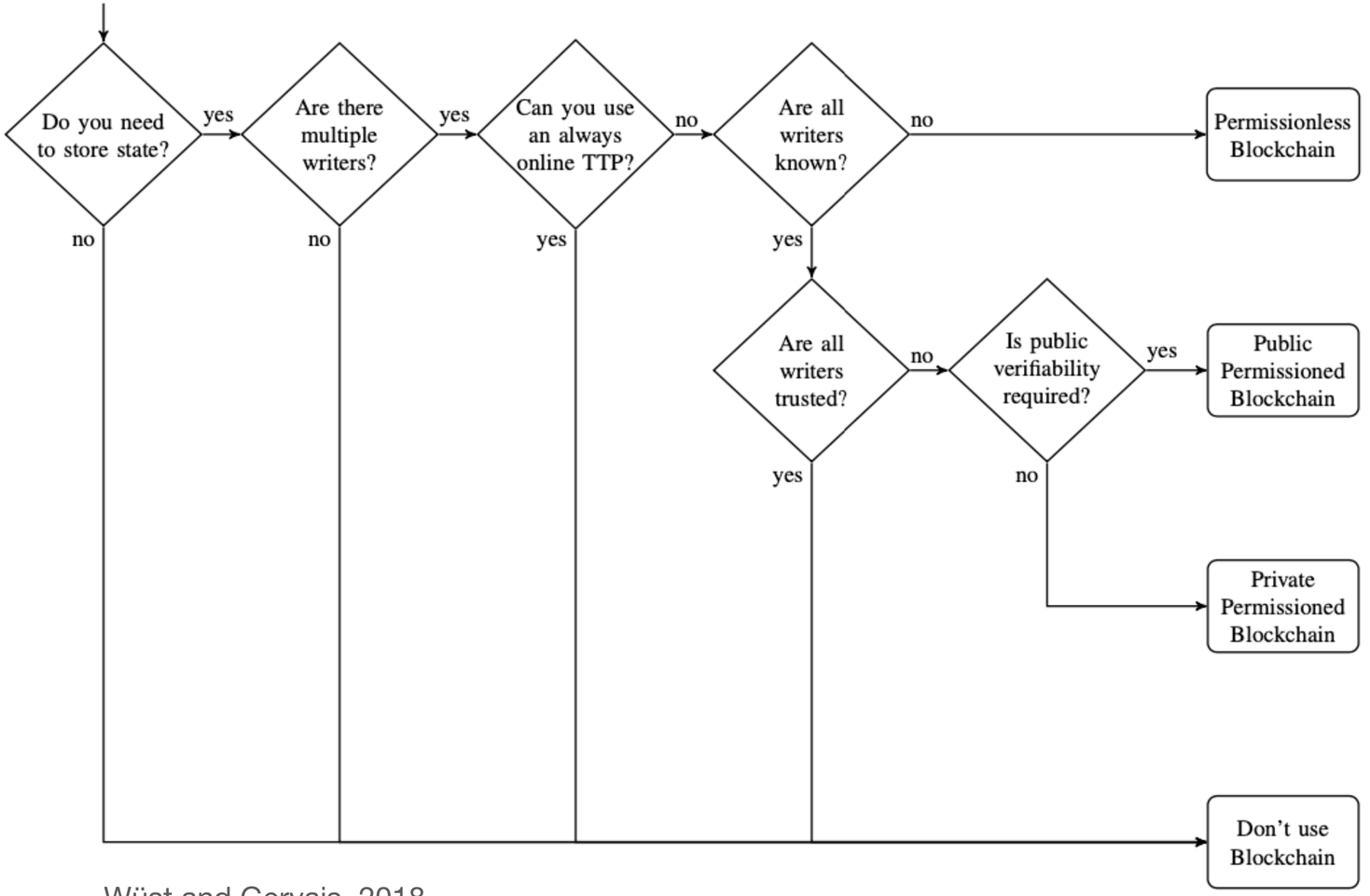
Takeaway 2

Options

- Single database
- Single organisation controlled blockchain
  - Gives: Resilience against attacks on individual machines
  - Careful to avoid common failure mode
- Multi organisation controlled blockchain
  - Transparency, Censorship resistance

Takeaway 3

Do you need a blockchain?



Wüst and Gervais, 2018